

Municipal Infrastructure Damage Reporting App

Requirements Table — Functional & Nonfunctional Requirements

Problem Statement #29 | Smart Cities, Transport & Logistics

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint the incident location on the municipal GIS map.	High	Pass: Pothole report is routed to the Road Infrastructure division with exact GPS coordinates. Fail: Report is submitted without location data.	Given (sample requirement).
FR-002	Functional	The system shall allow a citizen to categorize a damage report as Pothole, Broken Streetlight, or Water Pipeline Leak, and attach at least one geotagged photo before submission.	High	Pass: Submission is blocked with a validation message if no category or photo is attached; a report with both is accepted and assigned a ticket number within 5 seconds. Fail: A report is created without a category or without any photo attached.	Correct classification is required before a ticket can be routed to the right department.
FR-003	Functional	The system shall automatically dispatch a new ticket to the municipal engineering department mapped to the report's damage category (e.g., Road Infrastructure, Electrical/Streetlighting, Water Board) within 2 minutes of submission.	High	Pass: A test pothole report generates a ticket visible in the Road Infrastructure division's queue within 2 minutes, with category, photo and GPS coordinates attached. Fail: A submitted report has no ticket generated, or is routed to the wrong department.	Automated routing removes manual triage delay and ensures reports reach the responsible department without staff intervention.
FR-004	Functional	The system shall allow a citizen to track the real-time status (Submitted, Acknowledged, In Progress, Resolved) of any report they have filed, referenced by a unique ticket ID.	Medium	Pass: Entering a valid ticket ID displays the current status and last-updated timestamp. Fail: Status shown does not match the department's latest update, or an invalid ticket ID returns no error message.	Citizens need visibility into report progress to build trust in the municipal reporting process.
FR-005	Functional	The system shall allow a Municipal Engineer to view assigned tickets on a GIS map, update a ticket's status, attach resolution notes/photos, and close the ticket once work is complete.	High	Pass: An engineer changes a ticket from 'In Progress' to 'Resolved' with a note, and the citizen's tracked status updates accordingly. Fail: A status change made by the engineer is not reflected on the citizen-facing tracker.	Engineers need a single workflow to manage, resolve and formally close incoming tickets.
NFR-001	Nonfunctional	The reporting portal shall support offline report drafting and auto-sync when mobile network connectivity resumes.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given (sample requirement).
NFR-002	Nonfunctional	The system shall encrypt all uploaded photos and GPS metadata in transit (TLS 1.2+) and at rest (AES-256), and restrict ticket-management access to authenticated municipal engineer accounts.	High	Pass: A penetration/security scan confirms all data-in-transit uses TLS 1.2+ and stored photos are AES-256 encrypted; an unauthenticated request to the engineer dashboard is rejected with a 401/403 response. Fail: Any	Citizen-submitted photos and locations are personal data; the portal must protect them and prevent unauthorized ticket manipulation.

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				unencrypted transmission or storage of citizen photo/location data is detected, or an unauthenticated user can access the engineer dashboard.	